

[< Corn](#)

AGRICULTURE

Relative Toxicities of Insecticides and Miticides Used in Corn to Natural Enemies and Honey Bees

Updated: 08/2008



On this page



The Honey Bee Precaution Pesticide Ratings may differ from the pollinator precautionary statements on pesticide labels. These ratings come from the [UC IPM Bee Precaution Pesticide Ratings](#) and are developed independently from the *Pest Management Guidelines*. They are meant to provide supplementary information to help readers choose pesticides less hazardous to honey bee. For information about how these ratings are established, see [Bee Precaution Pesticide Ratings Frequently Asked Questions](#). Users of pesticides must always comply with all label instructions and legal

requirements, which represent the minimum precautions to protect honey bee.



 [Expand table](#)

Common name 	Example trade name
Bacillus thuringiensis ssp. aizawai	various products #
Bacillus thuringiensis ssp. kurstaki	various products #
carbaryl (bait)	Sevin 5 Bait
carbaryl—XLR formulation *	Sevin XLR Plus
clothianidin (seed treatment)	Poncho 600
dimethoate	Dimethoate 400
endosulfan *	Thionex 3EC
esfenvalerate	Asana XL
indoxacarb	Avaunt
malathion	Malathion 8-E
methomyl *	Lannate SP
methoxyfenozide	Intrepid 2F
mineral oil	various products #
permethrin	Ambush 25W
propargite	Comite
spinetoram	Radiant SC
spinosad	Entrust #

Common name	Example trade name
spiromesifen	Oberon 2 SC
thiamethoxam (seed treatment)	Cruiser 5FS

— No information.

I Do not apply or allow to drift to plants that are flowering including weeds. Do not allow pesticide to contaminate water accessible to bees including puddles.

II Do not apply or allow to drift to plants that are flowering including weeds, except when the application is made between sunset and midnight if allowed by the pesticide label and regulations. Do not allow pesticide to contaminate water accessible to bees including puddles.

III No bee precaution, except when required by the pesticide label or regulations.

- * [a b c](#) Permit required from county agricultural commissioner for purchase or use.
- # [a b c d](#) Acceptable for use on certified organic crops. Check with your certifier to confirm before application.
- ¹ [↔](#) Group numbers for insecticides and miticides are assigned by the [Insecticide Resistance Action Committee](#) (IRAC). Insecticides with unknown modes of action are assigned mode-of-action group numbers (MoAs) that begin with UN. Rotate pesticides with a different mode-of-action group number, and do not use products with the same mode-of-action group number more than twice per season to help prevent the development of resistance. For example, the organophosphates have a group number of 1B; insecticides with a 1B group number should be alternated with insecticides that have a group number other than 1B.

- ² [↩](#) Range of insect and mite groups affected by a pesticide.
Broad means the pesticide affects most groups of insects and mites; **narrow** means the pesticide affects only a few specific groups.
- ³ [↩](#) Risk of harm to honey bees. For more information, see [Bee Precaution Pesticide Ratings](#).
- ⁴ [↩](#) Risk of harm to predatory mites. Toxicities are generally to western predatory mite, *Galendromus occidentalis*. Where differences have been measured in toxicity of the pesticide-resistant strain versus the native strain, these are listed as pesticide-resistant strain or native strain.
- ⁵ [a b](#) Risk of harm to parasitoids and general predators. Toxicities are averages of reported effects and should be used only as a general guide. Actual toxicity of a specific insecticide depends on factors including the application rate, environmental conditions, and the life stage and species of a parasitoid or predator.
- ⁶ [↩](#) Length of time residue affects natural enemies. **Short** means hours to days; **moderate** means days to 2 weeks; and **long** means many weeks or months.
- ⁷ [↩](#) Date information was last updated in the *UC IPM Pest Management Guidelines*.
- ⁸ [↩](#) Use lowest rates for best management of predatory mite to spider mite ratio.
- ⁹ [a b c](#) Toxic to some natural enemies (lacewing and syrphid fly larvae, predatory beetles, and thrips) when sprayed and up to 5 to 7 days after.
- ¹⁰ [↩](#) Residual is moderate if solution is between a pH of 7 and 8.



UC IPM Pest Management Guidelines: Corn

UC ANR Publication 3443

[Authors and Credits](#)

Managing Pests

About

Agriculture

About UC IPM

Home and Landscape

Annual Reports

Natural Area Pests

UC IPM Strategic Plan

Pesticide Safety

UC IPM Programs

Invasive and Exotic Pests

What is IPM?

What's New

Resources

Get In Touch

Glossary

Contact Us

Links

Feedback

Pesticide Information

Publications and Newsletter

Recursos en español

Trainings and Events



UNIVERSITY OF CALIFORNIA
Agriculture and Natural Resources

Statewide Integrated
Pest Management Program



Home and Landscape: [Instagram](#) | [Facebook](#)

[Use Permissions and Citations](#) | [Nondiscrimination Statement](#) |
[Acknowledgements](#) | [Accessibility](#) | [Privacy Statement](#) | [Staff-Only
Page](#)

©1996—2026 Statewide IPM Program, Agriculture and Natural Resources,
University of California Regents of the University of California unless otherwise
noted.